



The Moon — FULL MIND MAP

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Chapter 01_xi_The_Moon

Topic Index

1. Apollo 11 — First Moon Landing
2. Moon ki Rotation — Synchronous
3. Moon — No Atmosphere
4. Lunar Eclipse
5. Moon ki Distance aur Brightness
6. Moon = Satellite, Planet Nahi
7. Blue Moon aur Weight Change
8. Moon Missions

Apollo 11 — First Moon Landing

- Human ne pehli baar Moon pe kadam rakha **20 July 1969** ko — Apollo 11 mission (astronauts: Neil Armstrong, Michael Collins, Edwin 'Buzz' Aldrin). [Q1]
- Armstrong aur Aldrin surface pe utre, Collins command module mein orbit karte rahe. Landing site: '**Sea of Tranquility**' (lunar mare Moon pe hota hai). [Q1, Q2]

Moon ki Rotation — Synchronous

- Moon ka lagbhag wahi face hamesha Earth ki taraf rehta hai — kyunki Moon ka axis-rotation period aur Earth-orbit period dono **~27.3 days** hain (synchronous rotation). [Q3, Q4]
- Libration ki wajah se Earth se time ke saath Moon ki surface ka ~59% dikh jaata hai. [Q3]

Moon — No Atmosphere

- Moon pe do log ek dusre ki awaaz directly nahi sun sakte kyunki **Moon pe atmosphere nahi** hai (sound ko medium chahiye). [Q5]

Lunar Eclipse

- Lunar eclipse tab hota hai jab **Earth Sun aur Moon ke beech** aa jaati hai — sirf **Full Moon/Poornima** ke din possible. [Q6, Q7, Q8]

Moon ki Distance aur Brightness

- 22 Dec 1999 ki full-moon brightness ka main reason: Moon **Perigee** position mein tha (Earth ke sabse close point). [Q9]

- Jab Moon half-illuminated dikhta hai, Sun-Earth-Moon ke beech angle **90°** hota hai. [Q10]
- Moon ki Earth se average distance ~**384 thousand km (384,400 km)** hai. [Q16]

■ Moon = Satellite, Planet Nahi

- Mars, Mercury, Earth planets hain; **Moon Earth ka satellite** hai (odd one out / not a planet). [Q11, Q12]
- Match: Planet-Uranus, Satellite-Moon, Comet-Halley, Artificial spacecraft-Mariner. [Q13]

■ Blue Moon aur Weight Change

- '**Blue Moon**' = same calendar month mein 2 full moons occur karna (2nd full moon ko Blue Moon kehte hain). [Q14]
- Earth se Moon pe object le jaane pe weight **kam ho jaata hai** (~1/6th) — Moon ka gravitational acceleration Earth ka ~1/6 hai; mass same rehta hai. [Q15]

■ Moon Missions

- **Japan** pehla Asian country tha jisne **SELENE (Kaguya)** lunar orbiter Moon ki orbit mein bheja (launched 14 Sept 2007, ~1 year 10 months orbit ki). [Q17]